

**UCSC****University of Colombo, Sri Lanka***University of Colombo School of Computing***DEGREE OF BACHELOR OF INFORMATION TECHNOLOGY
(EXTERNAL)**Academic Year 2023 — 3rd Year Examination — Semester 6**IT6405 (R) — Database Systems II
(Repeat Paper)***Structured Question Paper*
(2 Hours)**To be completed by the candidate****Index Number**

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Important Instructions

- The duration of the paper is **2 hours**.
- The medium of instructions and questions is English. Students should answer in the medium of English language only.
- This paper has **4 questions** on **12 pages**. Answer **all** questions.
- All questions carry **equal** marks.
- Write your answers **only on the space provided** on this question paper.
- Do not tear off any part of this question paper. Under no circumstances may this paper (or any part of this paper), used or unused, be removed from the Examination Hall by a candidate.
- Note that questions appear on both sides of the paper. If a page or part of a page is not printed, please inform the supervisor/invigilator immediately.
- Any electronic device capable of storing and retrieving text, including electronic dictionaries, smartwatches, and mobile phones, is not allowed.
- Calculators are **allowed**.
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**To be completed by
the examiners**

1	
2	
3	
4	
Total	

- 1) (a) Write the SQL query to create the Department table with the following constraints.

Department (deptId, deptName, location, deptHead, numEmployees)

- deptId is a number used as the primary key
- deptName cannot be null
- Default location to be 'Galle'
- deptHead specifies a unique number
- numEmployees should be a number between 1-50

[4 marks]

ANSWER IN THIS BOX

```
CREATE TABLE Department (
    deptId int Primary Key,
    deptName CHAR(20) NOT NULL,
    location CHAR(25) DEFAULT 'Galle',
    deptHead INT UNIQUE,
    numEmployees INT CHECK (numEmployees BETWEEN 1 AND 50));
```

- (b) Consider the following relational schema to answer the given question.

Employee (eno, ename, hiredate, salary, saldifference)

Write a trigger to calculate and update the saldifference filed (difference between the old and new salary) when the salary is updated.

[6 marks]

ANSWER IN THIS BOX

```
DELIMITER //
CREATE TRIGGER SalUpdate
AFTER UPDATE ON Employee
FOR EACH ROW
BEGIN
    DECLARE sal_dif INT;

    IF NEW.salary > OLD.salary THEN
        SET sal_dif = NEW.salary - OLD.salary;
    ELSE
        SET sal_dif = OLD.salary - NEW.salary;
    END IF;

    UPDATE Employee SET saldifference = sal_dif WHERE eno =
NEW.eno;
END//
```

DELIMITER ;

Similar correct answer.

- (c) Write down two (02) common uses of database triggers.

[4 marks]

ANSWER IN THIS BOX

- Provide sophisticated auditing
- Prevent invalid transactions
- Enforce referential integrity (either those actions not supported by declarative constraints or across nodes in a distributed database)
- Enforce complex business rules
- Enforce complex security authorizations
- Provide transparent event logging
- Automatically generate derived column values
- Enable building complex views that are updatable
- Track database events

Any two of the above.

- (d) Suppose there is a relation declared as follows:

```
CREATE TABLE Employee (Name VARCHAR(50) PRIMARY KEY,
Salary INT CHECK(Salary <= 40000) );
```

Initially, the relation has three records:

Name	Salary
Perera	10000
Costa	20000
Silva	30000

We execute the following sequence of modifications.

- (i) INSERT INTO Employee VALUES ('Dias', 12000);
- (ii) UPDATE Employee SET Salary = 50000 WHERE Name = 'Silva';
- (iii) INSERT INTO Employee VALUES ('Perera', 13000);
- (iv) DELETE FROM Employee WHERE Name = 'Costa';

At the end of these statements, calculate the sum of the Salaries over all the tuples in the Employee relation.

[4 marks]

ANSWER IN THIS BOX

52000

- (e) Consider the tables Employee and Evaluation as given below to keep track of employees, and their job evaluation records.

Employee (Empid, First_name, Job_id, Salary, Department_id)

Evaluation (Empid, Evaluation_date , Job_id, Department_id, Score)

Write down a trigger named **Salary_Increment** to increase salary of an employee as given below (based on the Score value) upon insertion of his/her evaluation record into the Evaluation table.

- 20% salary increment if score is ≥ 8

- 10% salary increment if score ≥ 6 and score < 8

[7 marks]

ANSWER IN THIS BOX

DELIMITER \$\$

```
CREATE TRIGGER Salary_Increment
AFTER INSERT ON Evaluation
FOR EACH ROW
BEGIN
  IF (NEW.Score  $\geq 8$ ) THEN
    UPDATE Employee
    SET Salary = Salary * 1.20
    WHERE Empid = NEW.Empid;
  ELSEIF (NEW.Score  $\geq 6$  AND NEW.Score  $< 8$ ) THEN
    UPDATE Employee
    SET Salary = Salary * 1.10
    WHERE Empid = NEW.Empid;
  END IF;
END$$
```

DELIMITER ;

Similar correct answer.

- 2) (a) Suppose that we have an ordered file with $r = 30,000$ records stored in a disk with block size $B = 1024$ bytes. File records are of fixed size and unspanned, with record size $R = 100$ bytes.

- (i) Calculate the blocking factor (BFR).

[2 marks]

ANSWER IN THIS BOX

```
BFR =  $\lfloor 1024/100 \rfloor$ 
      = 10 records per block.
```

- (ii) How many block accesses are required to search a record in the data file using the binary search?

[2 marks]

ANSWER IN THIS BOX

The number of blocks needed for the file is $b = \lceil (r/bfr) \rceil$
 $= \lceil (30,000/10) \rceil = 3000$ blocks.
 A binary search on the data file would need approximately $\lceil \log_2 b \rceil$
 $= \lceil (\log_2 3000) \rceil = 12$ block accesses.

- (iii) Suppose that the ordering key field of the file is $V = 9$ bytes long, a block pointer is $P = 6$ bytes long, and we have constructed a primary index for the file. How many block accesses are required to search a record using the primary index?

[4 marks]

ANSWER IN THIS BOX

Size of index entry is $9+6 = 15$ bytes.
 Blocking factor for index $= \lfloor (1024/15) \rfloor = 68$
 Number of index entries is 3000
 The number of index blocks is $= \lceil (3000/68) \rceil = 45$
 To perform a binary search on index file would need $\lceil (\log_2 b) \rceil = \lceil (\log_2 45) \rceil = 6$ block accesses.
 Total number of block access $= 6 + 1$ block accesses $= 7$ block accesses

- (b) Consider the following relational schema.

Employee (EmpId, Name, Address, Gender, Marital_Status, Dept_Id, Salary)

What is the most suitable index structure to efficiently retrieve the details of employees based on their Marital_Status and Gender?

[4 marks]

ANSWER IN THIS BOX

Bitmap index

- (c) Write down the difference between the **Dense Index** and the **Sparse Index**.

[4 marks]

ANSWER IN THIS BOX

Dense Index - Dense index contains an index record for every search key value in the data file.

Sparse Index - Instead of pointing to each record in the main table, the index points to the records in the main table in a gap (index item points to a block).

- (d) Briefly explain the **index only strategy** technique using a suitable example.

[3 marks]

ANSWER IN THIS BOX

A technique used by databases to retrieve data using only the index, without having to access the actual table data.

Suitable Example

- (e) Concurrent executions of transactions can cause problems for the database. Briefly explain the **Lost Update problem** and **Temporary Update (Dirty Read) problem** that can occur due to the concurrent execution of transactions. Use appropriate examples to describe the two problems.

[6 marks]

ANSWER IN THIS BOX

Lost Update problem

Occurs when two different transactions are trying to update the same column on the same row within a database at the same time.

T1	T2
READ(X) $X = X - N$	
	READ (X) $X = X + M$
WRITE(X) READ(Y)	
	WRITE(X)

Temporary Update (Dirty Read) problem

Occurs when one transaction updates a database item and then the transaction fails for some reason. Meanwhile,

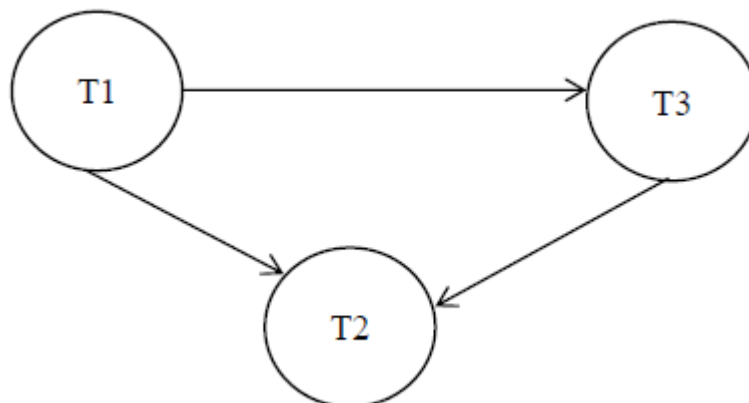
T1	T2
READ(X) $X = X - N$ WRITE(X)	
	READ (X) $X = X + M$ WRITE(X)
READ(Y) ROLLBACK	

- 3) (a) Consider the following schedule S, consisting of three transactions T1, T2, and T3. Note that each R and W denotes read, and write operations respectively.

T1	T2	T3
W(A)		
	R(A)	
W(B)		
		W(B)
		W(B)
	W(A)	
		R(B)
	R(B)	

- (i) Draw the precedence graph for S.

[5 marks]

ANSWER IN THIS BOX

- (ii) Is S conflict serializable? Justify your answer.

[3 marks]

ANSWER IN THIS BOX

Yes. Because the graph did not contain cycles.

$T1 \rightarrow T3 \rightarrow T2$

- (iii) Is S view serializable? Justify your answer.

[3 marks]

ANSWER IN THIS BOX

Yes. Since this is conflict serializable, this should be view serializable.

All the conflict serializable schedules are view serializable too.

- (b) Consider the following two schedules S1 and S2

S1: R1(X), R2(Y), W2(Y), R1(Y);

S2: W1(X), W2(X), W3(X), W1(Y), W2(Y);

- (i) Identify which schedule is allowed by Two-Phase Locking (2PL) and not by Time Stamp Ordering protocol

- (ii) Identify which schedule is allowed by Time Stamp Ordering and not by 2PL.

Note: R1(X) denotes Transaction 1 Read X value.

W2(X) denotes Transaction 2 Write X value.

[6 marks]

ANSWER IN THIS BOX

S1 is allowed by locks and not by timestamps.

Read X is allowed only if $t \geq w\text{-max}(X)$. Since $t1\text{-read}(B) < w\text{-max}(B)$, this condition is violated.

2PL can be applied

S2 is allowed by timestamps and not by locks.

Here it is not possible to acquire locks based on 2PL protocol.

- (c) Consider a table R (A) containing {(1), (2)}, and two transactions T1 and T2.
 T1: UPDATE R SET A = 2*A;
 T2: SELET AVG (A) FROM R;

If transaction T2 is executed using READ UNCOMMITTED, what would be the possible return values?

[4 marks]

ANSWER IN THIS BOX

1.5, 2, 2.5, 3

- (d) Briefly explain the properties of a transaction

[4 marks]

ANSWER IN THIS BOX

Atomicity: A transaction is an atomic unit of processing. It is either performed in its entirety or not performed at all.

Consistency preservation: A correct execution of the transaction must take the database from one consistent state to another.

Isolation: A transaction should not make its updates visible to other transactions until it is committed.

Durability (Permanency): Once a transaction changes the database and the changes are committed, these changes must never be lost because of subsequent failures.

- 4 (a) State two (02) advantages of distributed databases.

[4 marks]

ANSWER IN THIS BOX

Reflects organizational structure
 Local autonomy
 Improved availability
 Improved performance
 Modularity

- (b) State two (02) disadvantages of distributed databases.

[4 marks]

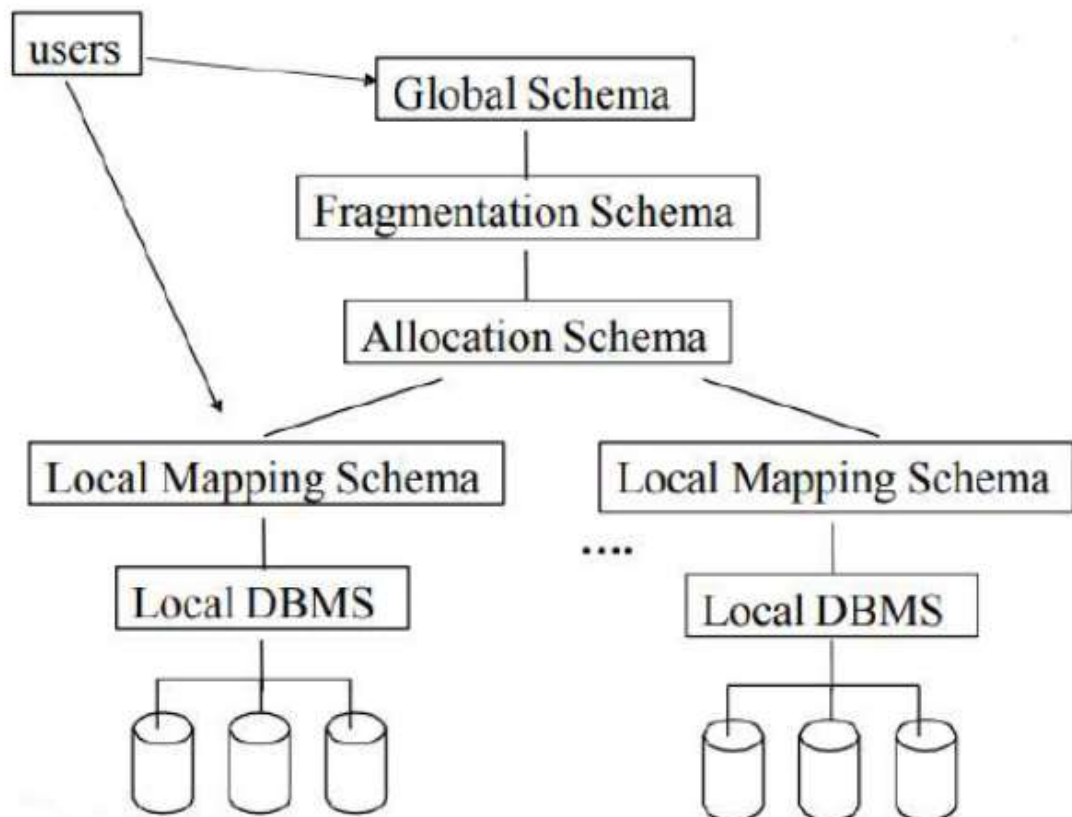
ANSWER IN THIS BOX

Complexity
Economics
Security
Difficult to maintain the integrity
Inexperience

- (c) Draw the architectural diagram for a distributed database management system and identify its components.

[7 marks]

ANSWER IN THIS BOX



- (d) Consider the following relational schema.

Employee (EmpId, Fname, Lname, Address, Gender, Salary, Bdate)

Project (Pnumber, Pname, location)

Works_on (Pno, Empid,)

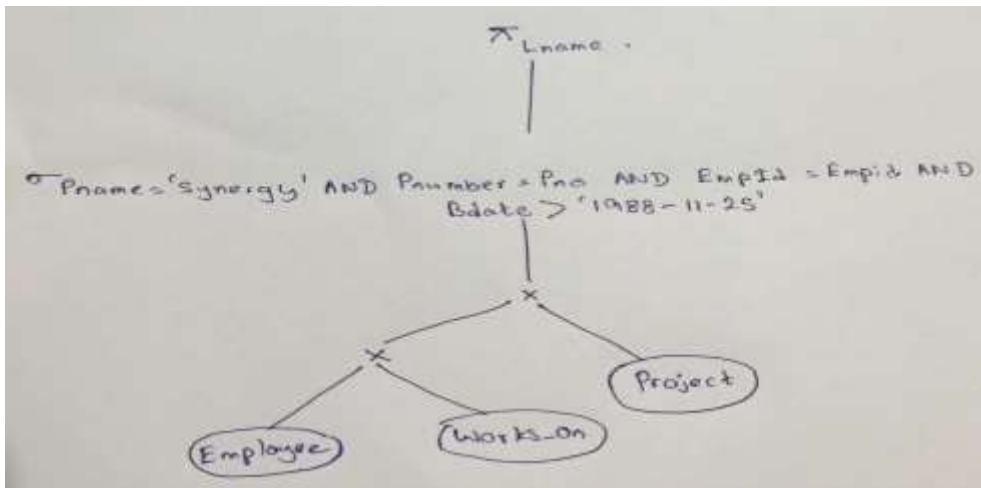
Consider the following SQL query.

```
Select E.Lname
From Employee E, Works_on W, Project P
Where P.Pname = 'Synergy' And
P.Pnumber= W.Pno And E.EmpId=W.Empid
And E.Bdate > '1988-11-25';
```

- (i) Draw the initial query tree for the SQL query given above.

[4 marks]

ANSWER IN THIS BOX



(ii)

Draw the optimized query tree for the SQL query given above.

[6 Marks]

